

EMCH 260

SOLID MECHANICS

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Stress

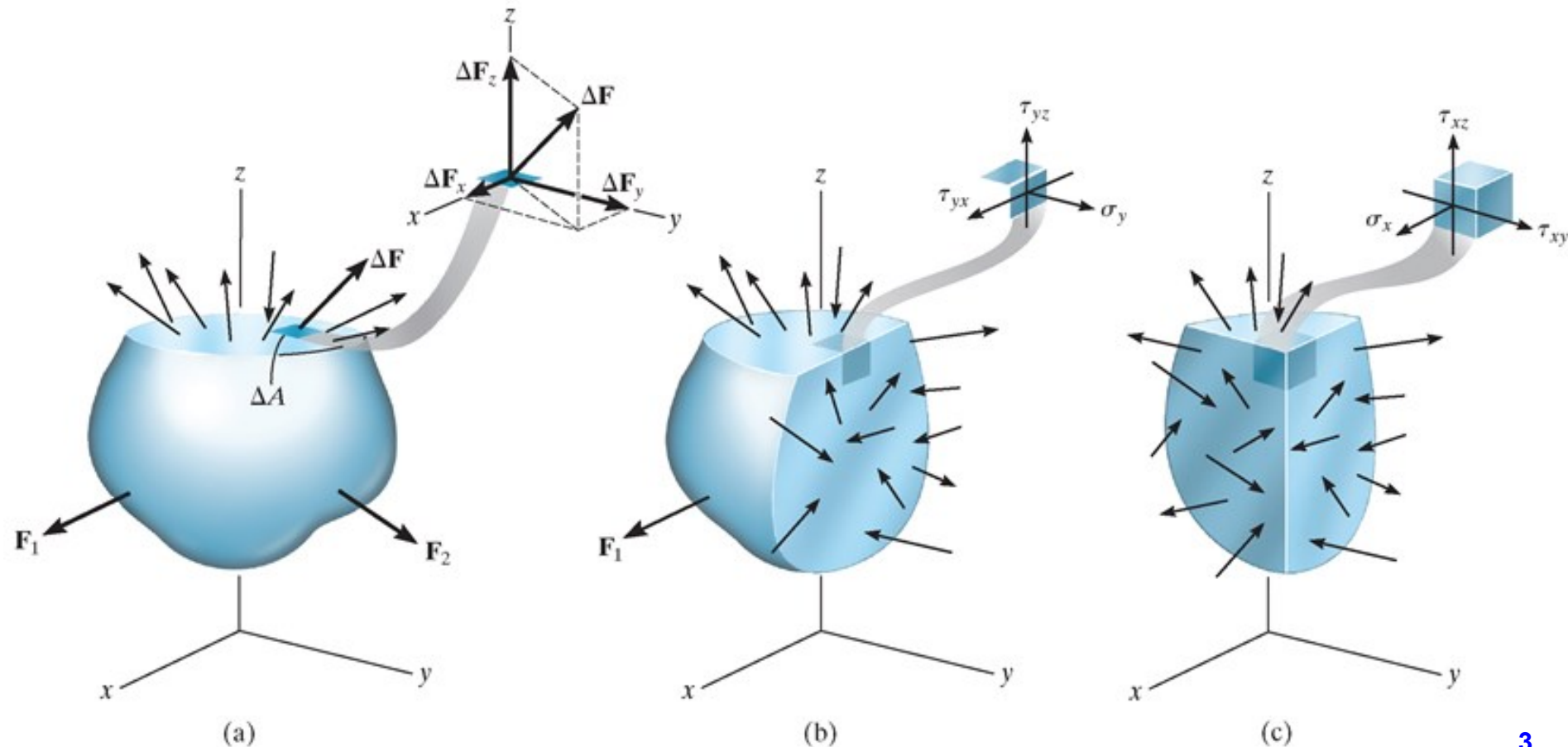
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Chapter 1.3

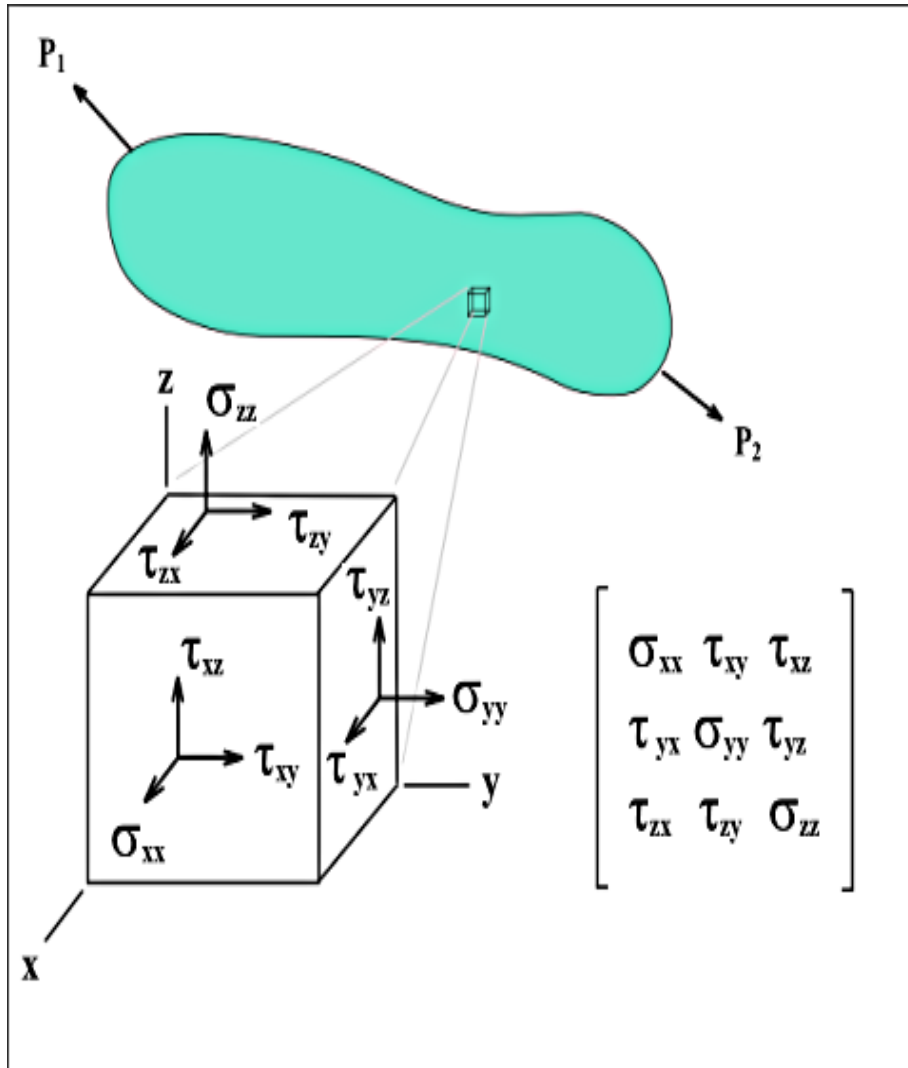
- ✓ Understand the concepts of average normal and shear stress
- ✓ Analyze and design of members subjected on axial load or shear
- ✓ Know how to do allowable stress design

STRESS – GENERAL DEFINITION

- For any macroscopically small area ΔA , there is a typical finite yet very small force ΔF . When using the continuum concept ($\Delta A \rightarrow 0$), $\Delta F \rightarrow 0$. But $\Delta F / \Delta A \rightarrow \text{finite limit}$ (**intensity of the internal force acting on a specific plane**).



STRESS – GENERAL DEFINITION, cont'd



- Stress:

$$\sigma_{ji} = \lim_{\Delta A_j \rightarrow 0} \frac{\Delta F_i}{\Delta A_j}$$

- ✓ In principle, stress is a second-order tensor with 9 components, but can be fully described with 6 components due to symmetry in the absence of body moments.
- ✓ In N dimensions, the stress tensor σ_{ij} is defined by:

$$dF_i = \sum_{j=1}^N \sigma_{ji} dA_j$$

- ✓ Since ΔA can be arbitrarily chosen, the cubic volume can be in any orientation and corresponding σ_{ij} is also different

NORMAL STRESS AND SHEAR STRESS

- Normal Stress (σ):** the force acting normal to ΔA

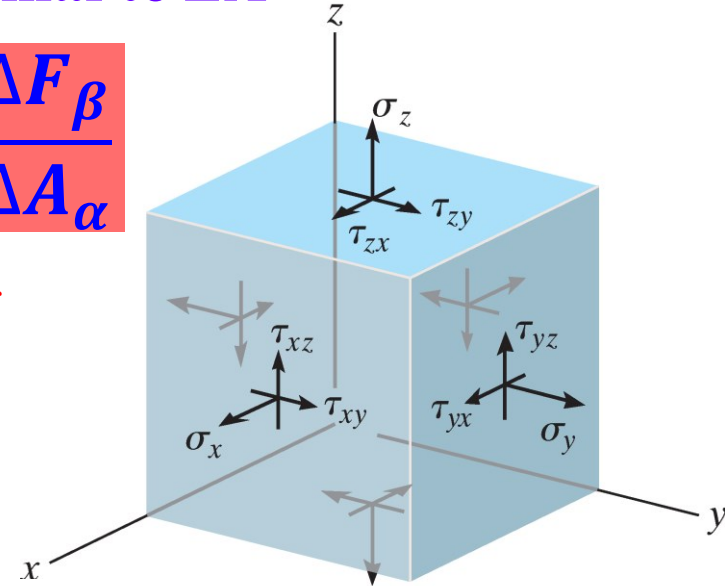
$$\sigma_x = \lim_{\Delta A \rightarrow 0} \frac{\Delta F_x}{\Delta A}$$

$$\sigma_y = \lim_{\Delta A \rightarrow 0} \frac{\Delta F_y}{\Delta A}$$

$$\sigma_z = \lim_{\Delta A \rightarrow 0} \frac{\Delta F_z}{\Delta A}$$

$$\sigma_{\alpha\beta} = \lim_{\Delta A_{\alpha} \rightarrow 0} \frac{\Delta F_{\beta}}{\Delta A_{\alpha}}$$

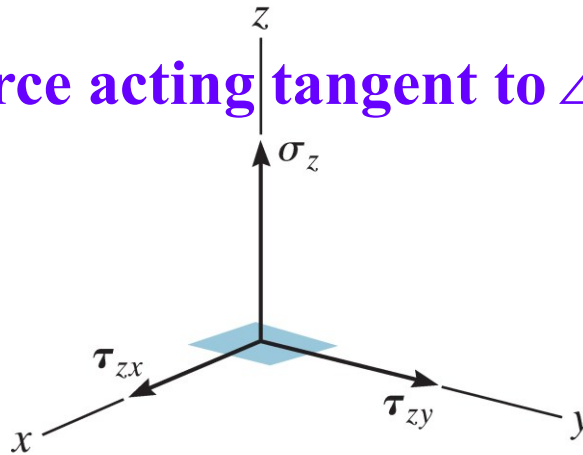
NOTE: ΔA is different for different normal stress



- Shear Stress (τ):** force acting tangent to ΔA

$$\tau_{zx} = \lim_{\Delta A_z \rightarrow 0} \frac{\Delta F_x}{\Delta A_z}$$

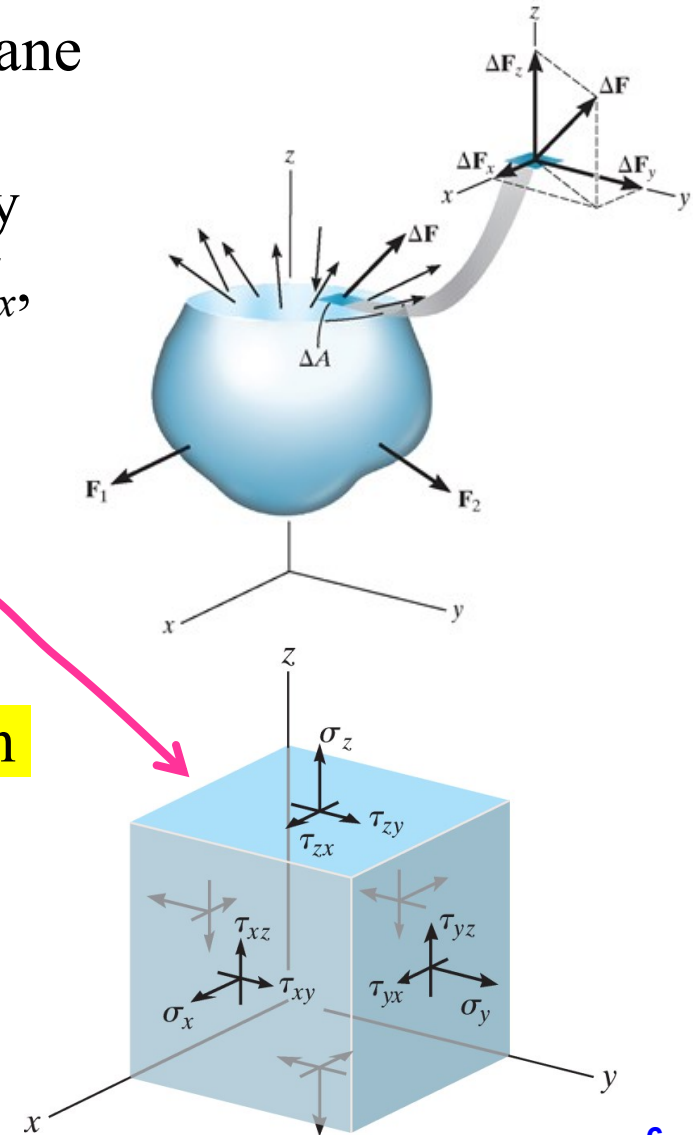
$$\tau_{zy} = \lim_{\Delta A_z \rightarrow 0} \frac{\Delta F_y}{\Delta A_z}$$



Units: N/m², Pa, pounds per square inch (psi)
(1 psi = 6.8947572932 kPa)

DISTINGUISH STRESS AND INTERNAL FORCE

- Internal force at any point or along any plane or perpendicular to any plane or in any orientation is **always a vector** and can only be in three directions with components (F_x , F_y , F_z)
- Stress is **always a tensor** and can be represented by a stress “cubic”
- Both stress and internal force are **direction dependent**
- Definition of stress is based on internal force, so they are strongly coupled



HOMOGENEOUS & ISOTROPIC VS. ANISOTROPIC

Isotropic: the material has the same physical and mechanical properties throughout its volume and in different directions

(single crystalline materials, fluids, e.g. Cu, Fe, water, air)

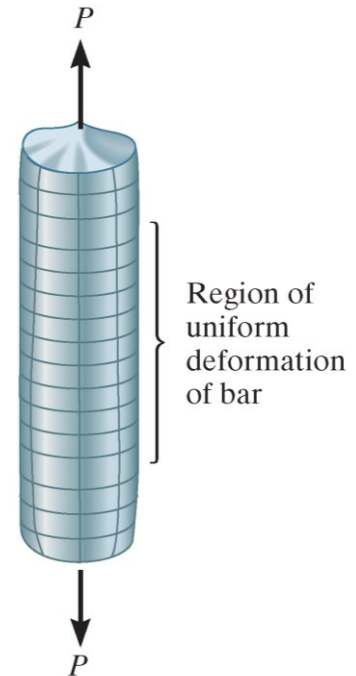
Anisotropic: the material has different properties in different directions

(most of amorphous materials, composites, e.g. glass, wood, food, clothes)



Homogeneous and isotropic assumption is needed to simplify the analysis of stress

For an axially loaded bar, the center region deforms uniformly and homogeneously



AVERAGE NORMAL STRESS

For an axially loaded bar, equation of equilibrium $\rightarrow$ *internal resultant normal force* $N = P$ (section taken perpendicular to the longitudinal axis of the bar).

Assumption: Homogeneous and isotropic material $\rightarrow$ **constant** normal stress distribution in the cross-section plane ($\Delta N = \sigma \Delta A$)

General equation

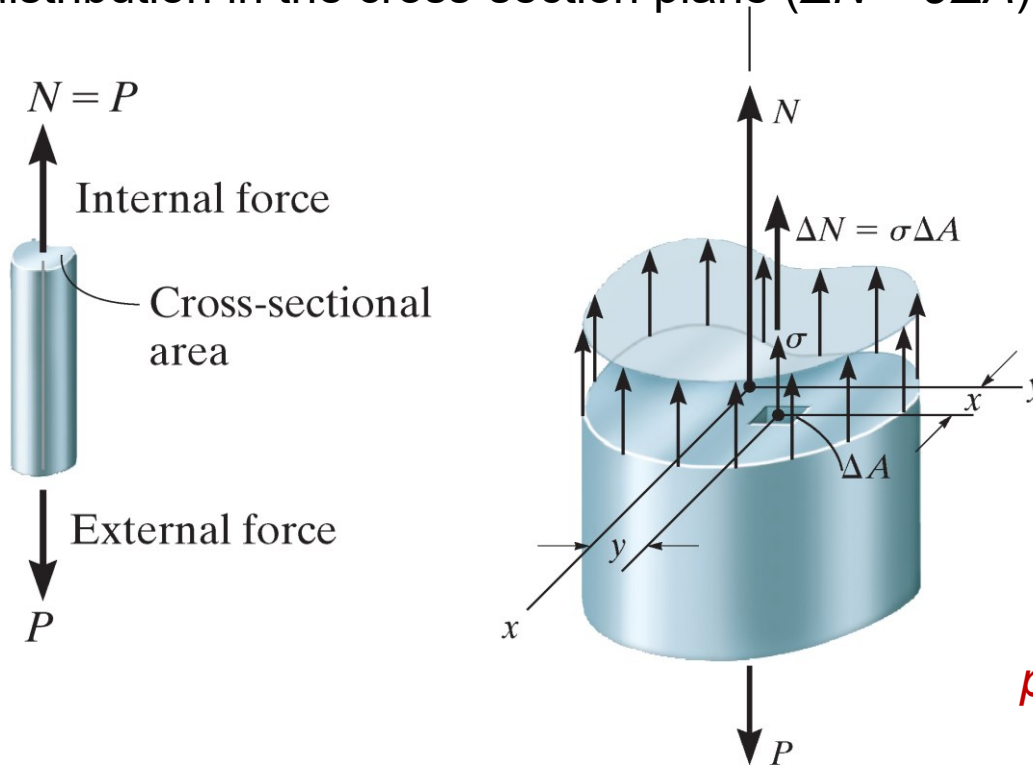
$$\int dN = \int_A \sigma dA$$

$$N = \sigma A$$

After
assumption

$$\sigma = \frac{N}{A}$$

(average **normal** stress at any point in the cross-sectional plane)

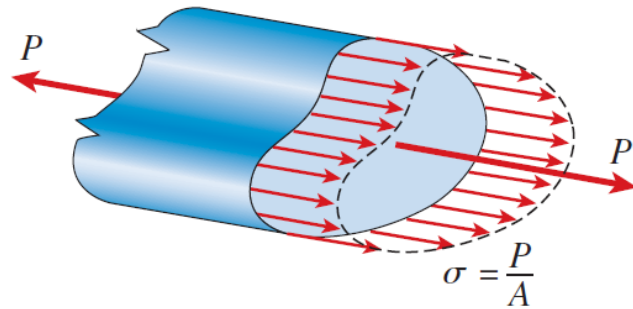


IMPORTANT: The magnitude of the stress at a point depends on the type of loading acting on the body and **the orientation of the element at the point**.

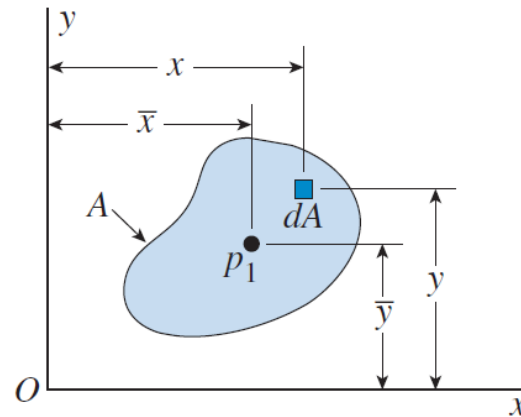
IMPORTANT POINTS OF NORMAL STRESS

Three conditions for well-defined normal stress:

- (1) Homogeneous and isotropic material
- (2) The load is along the longitudinal axis of the material
- (3) The “cut” section is perpendicular to the longitudinal axis



(a)



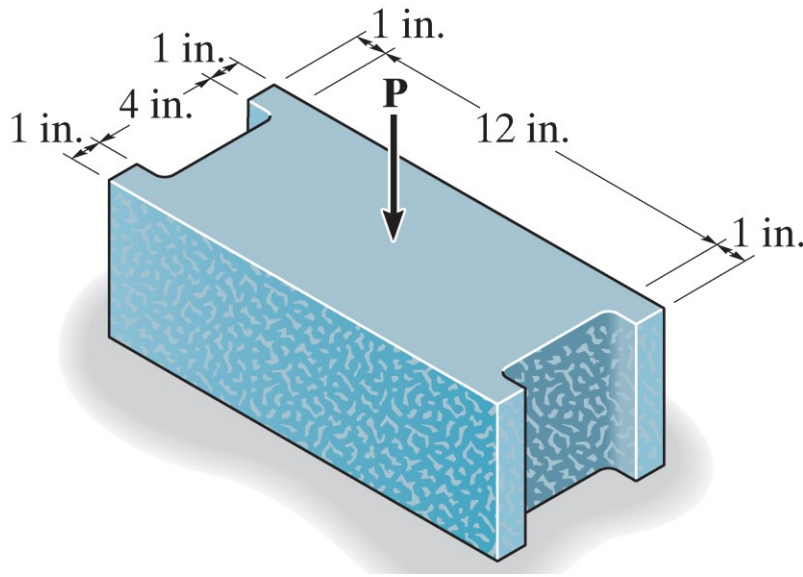
(b)

$$\sigma = \frac{P}{A}$$

Uniform stress distribution in a prismatic bar: (a) axial forces P , and
(b) cross section of the bar

IN-CLASS EXAMPLE 1: NORMAL STRESS

If the block is subject to a centrally applied force of $P = 6$ kip, determine the average normal stress in the material.



Solution:

Average Normal Stress: The cross-sectional area of the block is

$$A = 14 \times 6 - 2 \times 4 \times 1 = 76 \text{ in}^2$$

Thus,

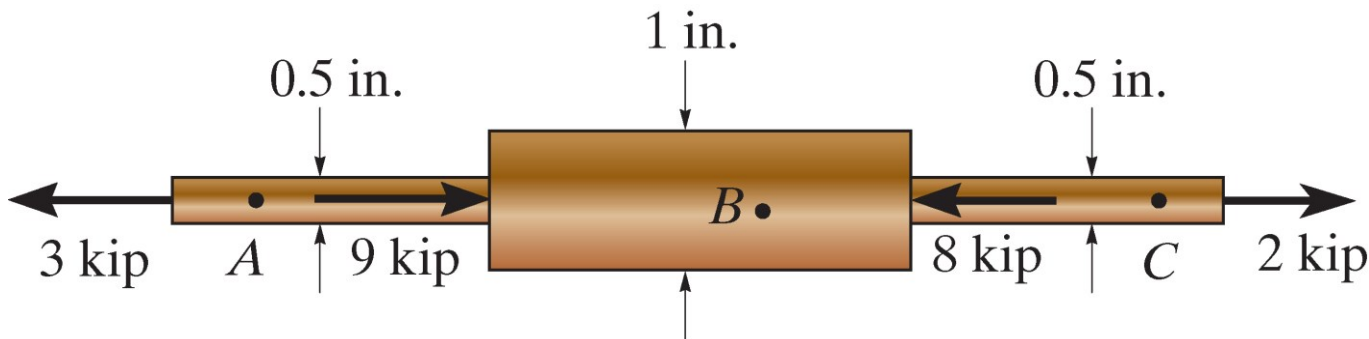
$$\begin{aligned}\sigma &= \frac{N}{A} \\ &= \frac{6 \times 10^3}{76} \\ &= 78.9 \text{ psi}\end{aligned}$$

MAXIMUM AVERAGE NORMAL STRESS

For cross-section changing materials or frames, different perpendicular cross-sections can be chosen (**note: $A_1, A_2, \dots, A_n$ could be different**)

$$\int dN = \int_{A_1} \sigma_1 dA_1 = \int_{A_2} \sigma_2 dA_2 = \dots = \int_{A_n} \sigma_n dA_n$$

Example: determine the average normal stress at points A, B, and C.



MAXIMUM AVERAGE NORMAL STRESS

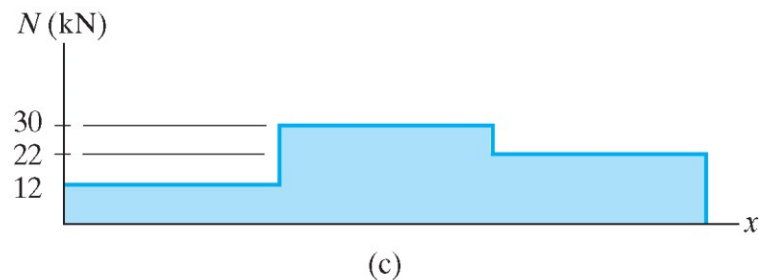
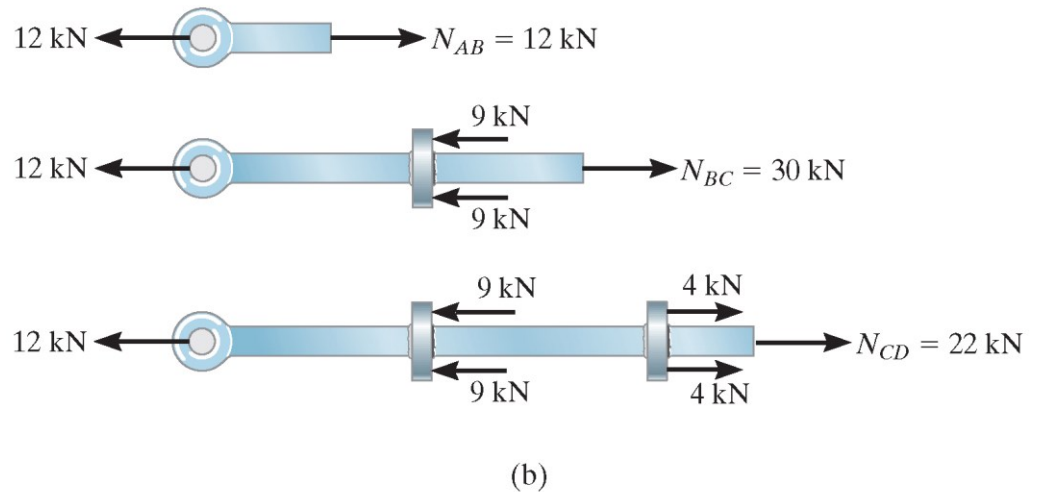
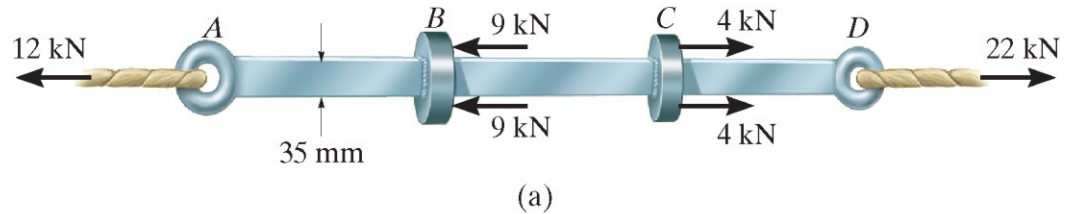
Or, for several external co-axial loads (note: $A_1, A_2, \dots, A_n$ could be different)

$$N_1 = \int_{A_1} \sigma_1 dA_1$$

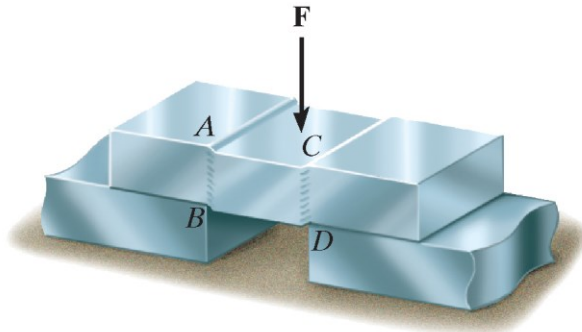
$$N_2 = \int_{A_2} \sigma_2 dA_2$$

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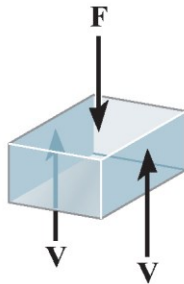
$$N_n = \int_{A_n} \sigma_n dA_n$$



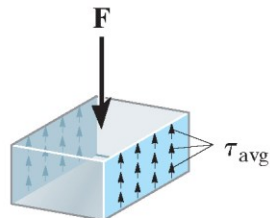
AVERAGE SHEAR STRESS



(a)



(b)



(c)

- Internal resultant shear force ($V = F/2$) is induced by the **relative movement of adjacent segments** upon applying external force (F)
- Average shear stress over the sectioned area (A), with same direction as V*

$$\tau_{avg} = \frac{V}{A}$$

- Analog to normal stress

General equation

$$\int dV = \int_A \tau dA$$

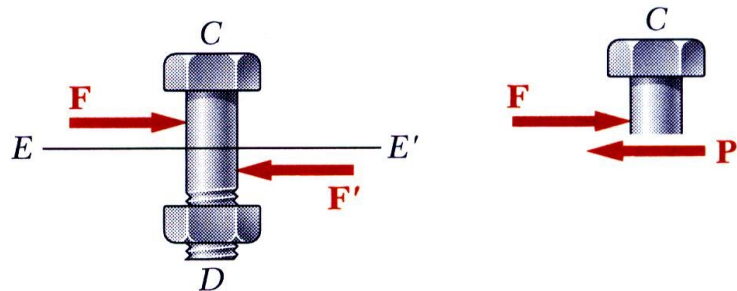
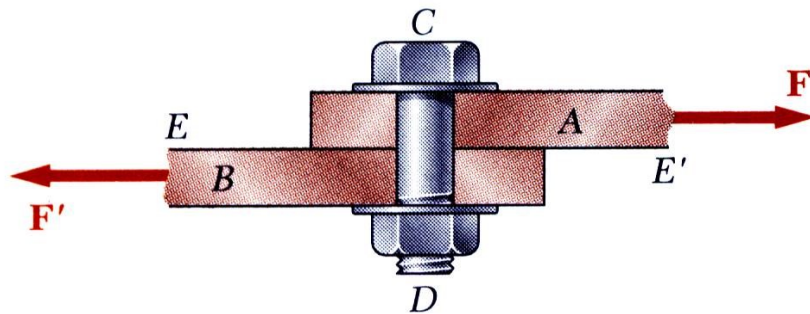
$$V = \tau A$$

After assumption

$$\tau = \frac{V}{A}$$

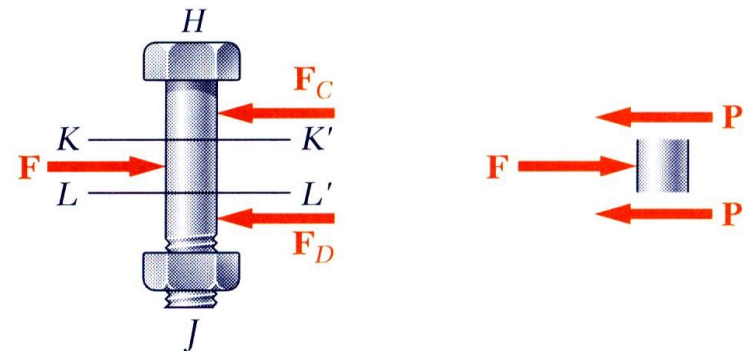
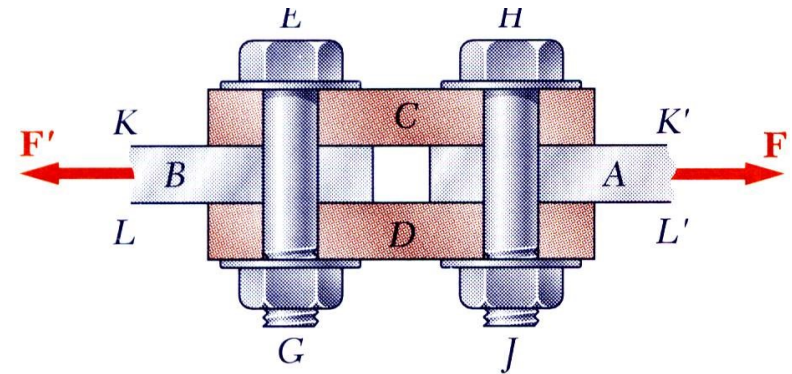
SHEARING STRESS EXAMPLES

Single Shear



$$\tau_{\text{ave}} = \frac{P}{A} = \frac{F}{A}$$

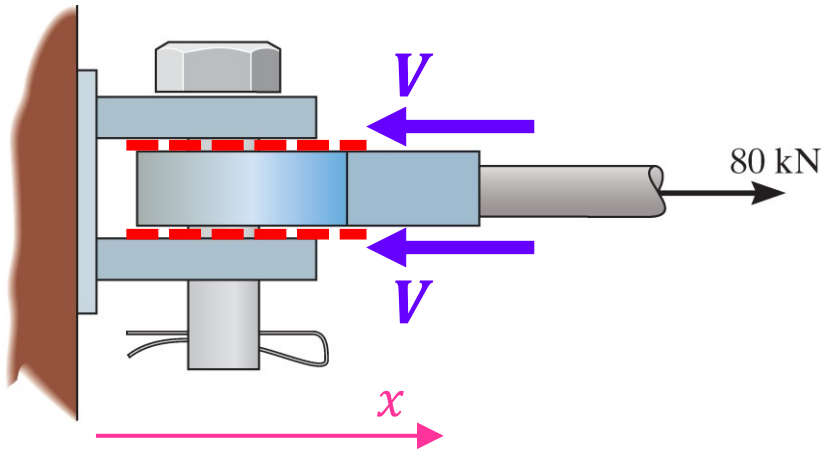
Double Shear



$$\tau_{\text{ave}} = \frac{P}{A} = \frac{F}{2A}$$

IN-CLASS EXAMPLE 2

Determine the shear stress in the pin. The diameter of the pin is 36 mm.



Solution:

The pin has double shear (red dashed lines).

Applying force equilibrium in x -direction:

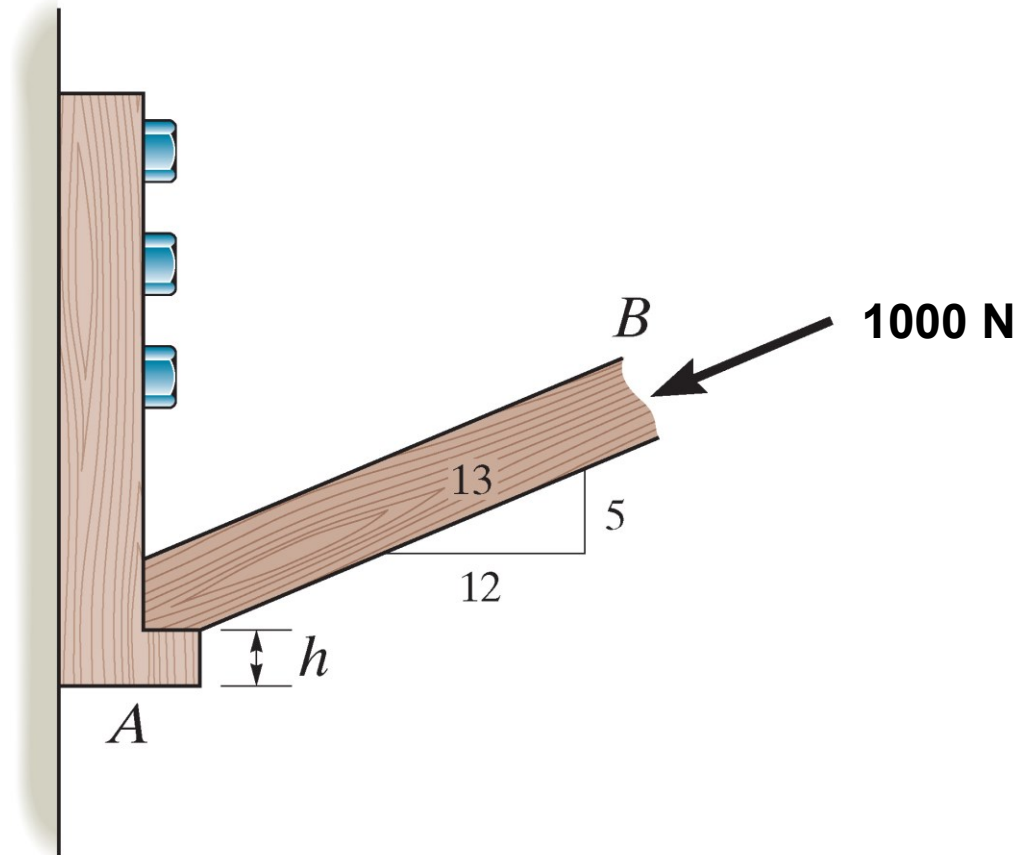
$$80 \text{ kN} - 2V = 0 \quad \rightarrow \quad V = 40 \text{ kN}$$

Then the shear stress at each sectioned plane is

$$\begin{aligned} \tau &= \frac{V}{\pi D^2 / 4} \\ &= \frac{40 \times 10^3}{3.1415926 \times (36 \times 10^{-3})^2 / 4} \\ &= 39.3 \text{ MPa} \end{aligned}$$

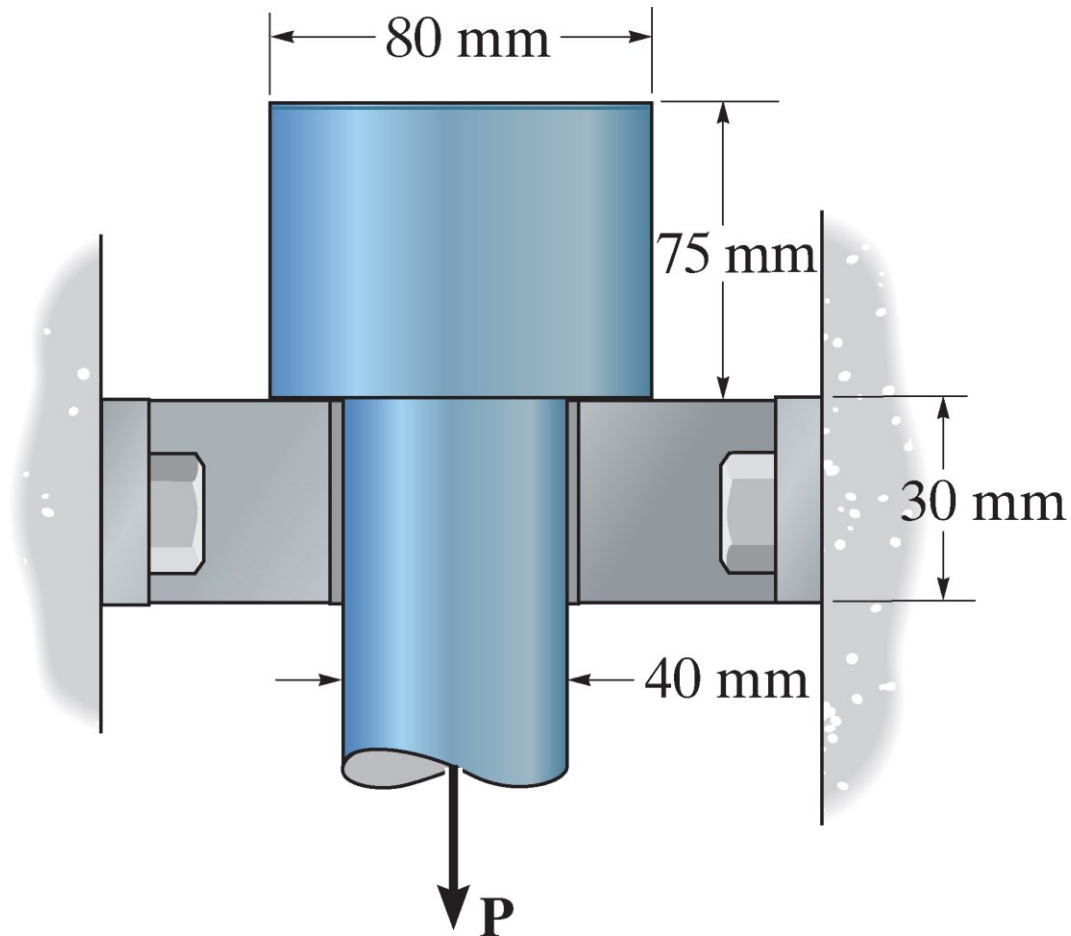
OUT-OF-CLASS THINK 1

Where the shear stress happens in these cases?



OUT-OF-CLASS THINK 2

Where the shear stress happens in these cases?



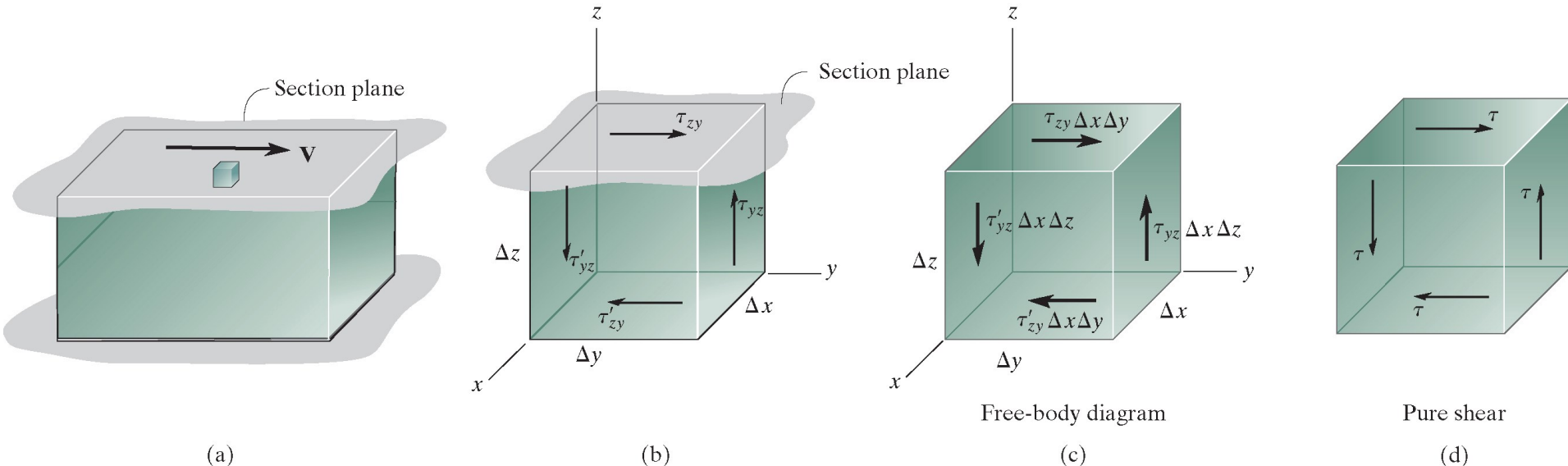
PURE SHEAR

- A shear force V on the surface of a cubic will result in shear stress on three other sides.

WHY we don't consider normal stress here?

- Force equilibrium: $\tau_{zy} = \tau'_{zy}$
 - Moment equilibrium: $\tau_{zy} = \tau_{yz}$
- $$\left. \begin{array}{l} \tau_{zy} = \tau'_{zy} \\ \tau_{zy} = \tau_{yz} \end{array} \right\} \tau_{zy} = \tau'_{zy} = \tau_{yz} = \tau'_{yz} = \tau$$

Complementary property of shear (pure shear)



RELEVANT PRACTICE

- Assignment 1, problem #2-3 (go to Blackboard under the “Assignments” folder, **Assignment 1 due: 8:30 AM, Monday, Sept. 13th 2021**)
- Optional Extra Exercise 1, problem #2 (go to Blackboard under the “Assignments” folder, **Assignment 1 due: 8:30 AM, Monday, Sept. 13th 2021**)
- *Submit your solutions directly onto Blackboard*

OUT-OF-CLASS READING

- **Textbook, pp. 22 – 37**